

5d) We use a proof by formula:

As $n \rightarrow \infty$ the regular polygon approaches a perfect circle. In a unit circle, for any point P at distance r from the center of the circle, the closest perimeter point Q is along the identical radial line.

Therefore, $PC = r$ and $PQ = 1 - r$

Setting $PC < PQ$ implies $r < 1 - r$ and that implies $r < 0.5$

Since the area of the circle scales r^2 , the area where this holds true is smaller of radius 0.5. Thus, the ratio of the area to the full circle is:

$\pi(0.5)^2 / \pi(1)^2 = 0.25$. This closely matches the output of the Monte Carlo simulation

for when we chose $n = 10$ million